



BUILD. CREATE. SCALE.

*Local Problems, Intelligent Solutions, Global
Impact to the World*

By

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**Will Africa merely
consume AI built
elsewhere, or will we
shape the future of AI?**

TECH SPEC: AI SOVEREIGNTY v. DEPENDENCE

GEO-DATA: AFRICAN MARKET DYNAMICS

TECH SPEC: AI SOVEREIGNTY v. DEPENDENCE

The Infrastructure Paradox

Distribution vs. Compute



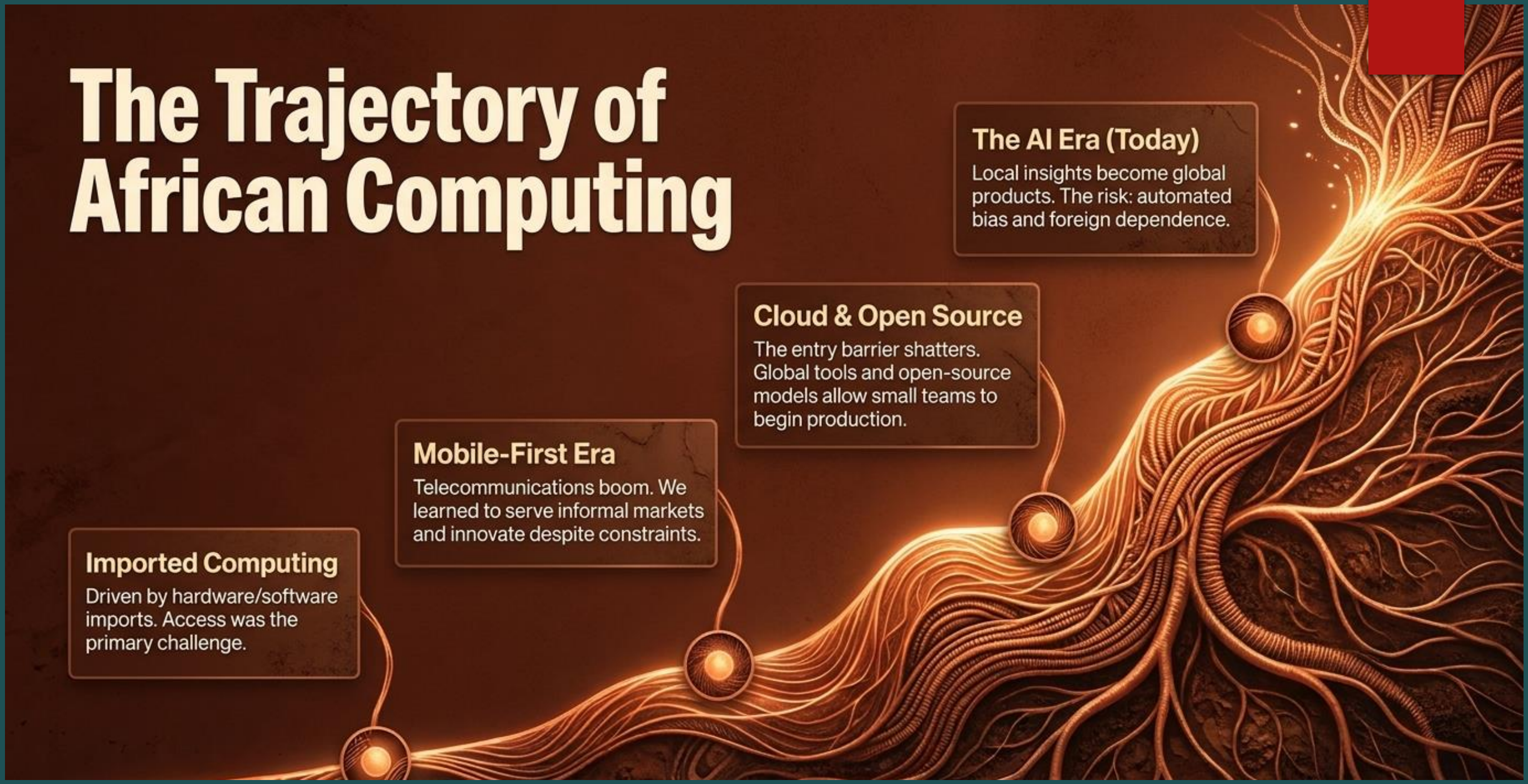
490M+

Sub-Saharan Mobile Subscribers



<1%
Global Compute
Capacity

The Trajectory of African Computing



Imported Computing

Driven by hardware/software imports. Access was the primary challenge.

Mobile-First Era

Telecommunications boom. We learned to serve informal markets and innovate despite constraints.

Cloud & Open Source

The entry barrier shatters. Global tools and open-source models allow small teams to begin production.

The AI Era (Today)

Local insights become global products. The risk: automated bias and foreign dependence.

AI Does Not Grow in Empty Space

TECH SPEC: LOCAL DATA AS FOUNDATIONAL INFRASTRUCTURE

**Representative Data
= Fair, Accurate AI**

**Imported Data
= Weak, Biased AI**

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Data

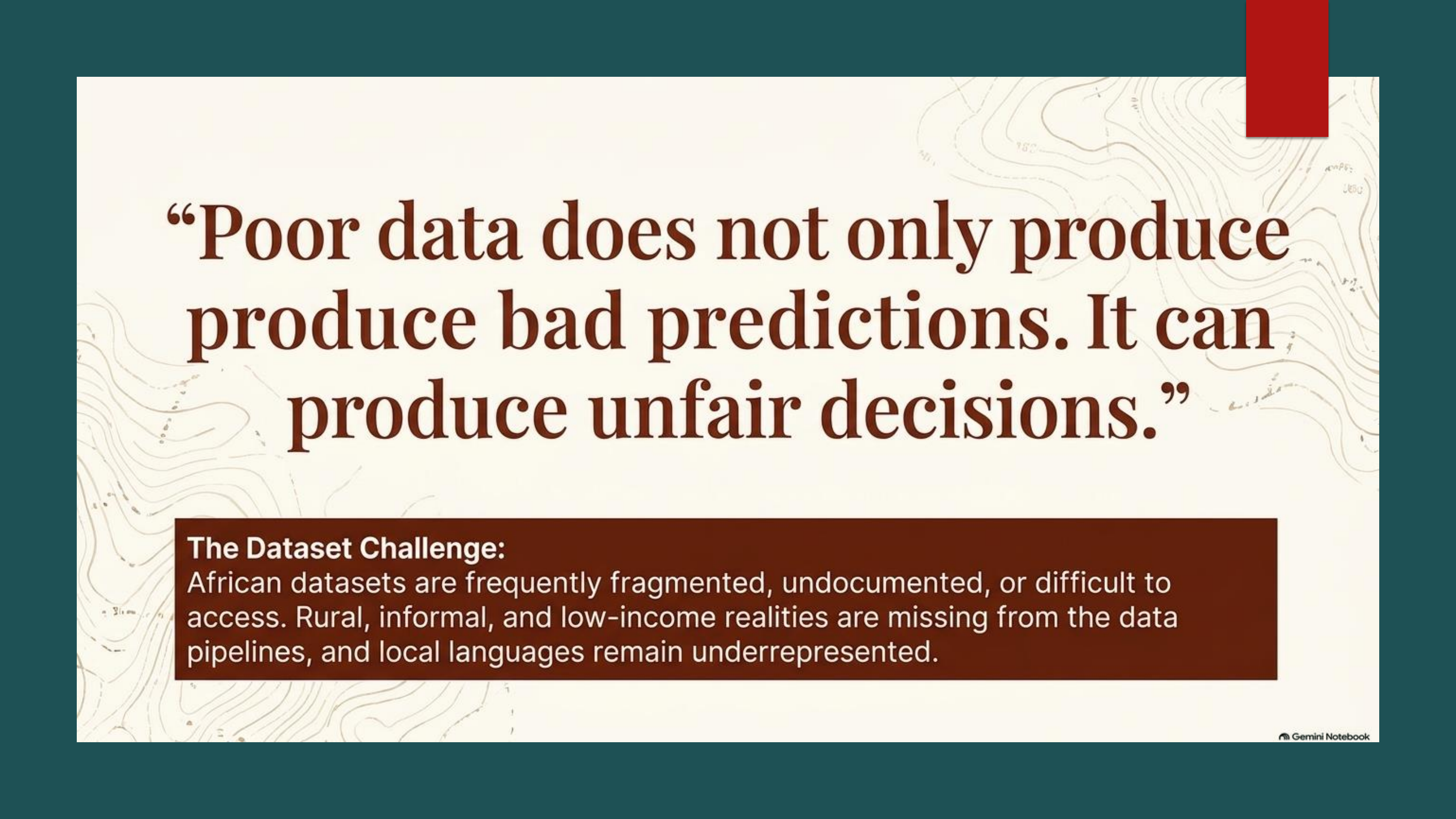
Health

Agriculture

Finance

Local Languages

Imported Datasets



“Poor data does not only produce
produce bad predictions. It can
produce unfair decisions.”

The Dataset Challenge:

African datasets are frequently fragmented, undocumented, or difficult to access. Rural, informal, and low-income realities are missing from the data pipelines, and local languages remain underrepresented.

The Anatomy of Representative Data

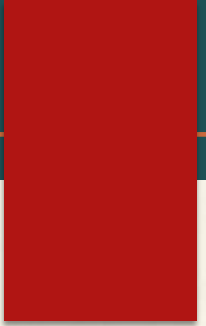


Wide Reach: Spans urban cities and rural villages equally.

Real Soil: Reflects **actual behaviours**, not idealised assumptions.

The Result: A resilient AI model that reflects the **people, languages,** and contexts where the system will actually operate.

Deep Roots: Captures diverse genders,



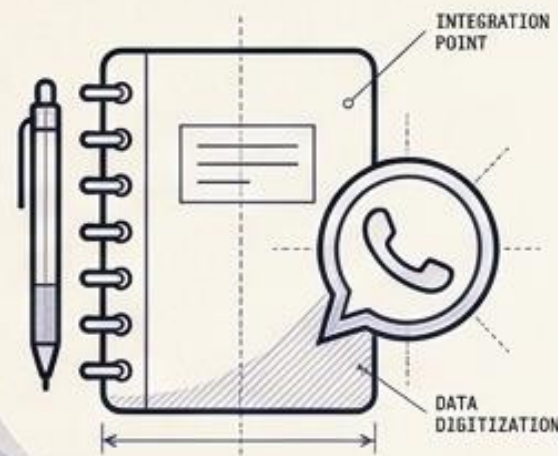
90%

3 in 10 die because nobody wanted what they built.

A free app nobody pays for isn't a startup. It's a hobby with servers.

Where to Find Your AI Blueprint

Stop looking for clever ideas. Look for deep frustrations.



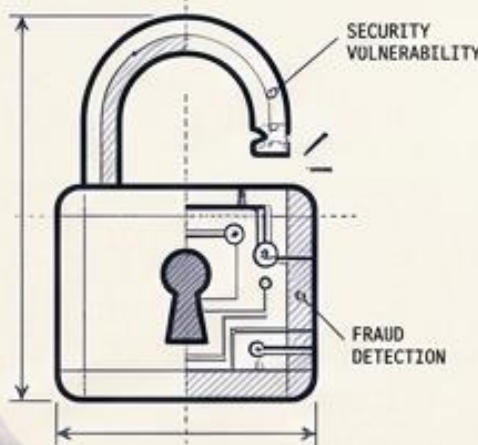
Offline Workflows

(WhatsApp + Notebooks)



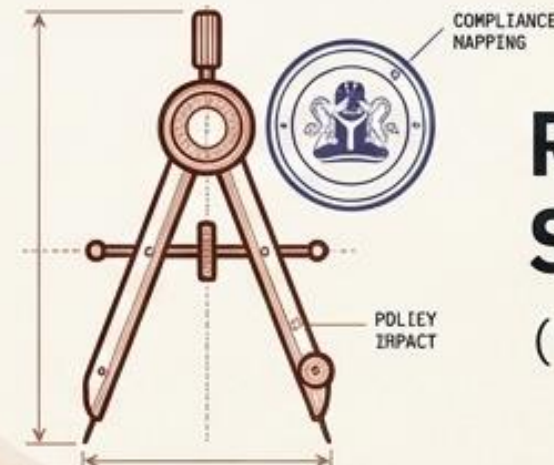
Personal Frustration

(Your day-to-day pain)



Broken Trust

(Where people are getting scammed)

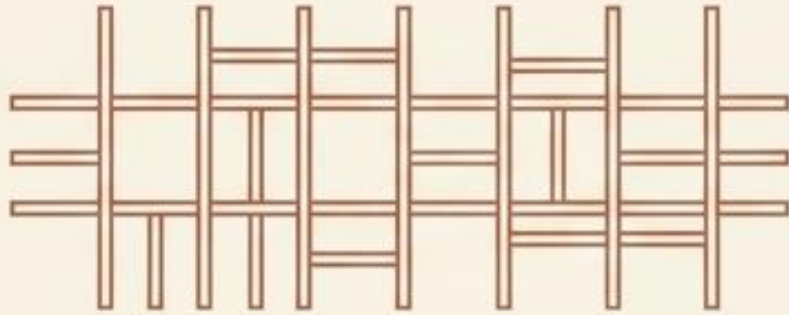


Regulatory Shifts

(New CBN/tax rules)

Language is Digital Infrastructure

True understanding requires context, not just translation.



Imported Assumptions

- One-size-fits-all defaults, dictionary translations, text-heavy interfaces.



Culturally Aware AI

- Understands local names and naming systems.
- Handles African accents, code-switching, and noisy environments.
- Respects community/family structures and informal market behaviours.
- Optimises for users more comfortable speaking than typing.

Engineering for Constraints

Design for real users on real networks, not laboratory conditions.

Constraint	Innovation
Unstable Power & Costly Data	Offline-first caching & low-bandwidth workflows.
Weak Network Connectivity	Edge AI and local model compression.
Low-End / Shared Devices	Voice-first, SMS, or USSD-supported architectures.
High-Stakes Environments (Health/Ag)	Human-assisted decision flows; keeping humans in the loop.

The Sector Impact Hub

Agriculture:
Mobile crop disease
detection,
weather-aware
planting, and yield
forecasting.



Healthcare:
Clinic triage support,
medical image
screening, and
disease outbreak
prediction.



African AI Ecosystem

Education:
Personalised local-
language tutors
and early warning
systems for
struggling students.



Fintech:
Fraud detection,
agent network
security, and credit
scoring for
informal traders.



Governance:
Citizen complaint
analysis, procurement
monitoring, and disaster



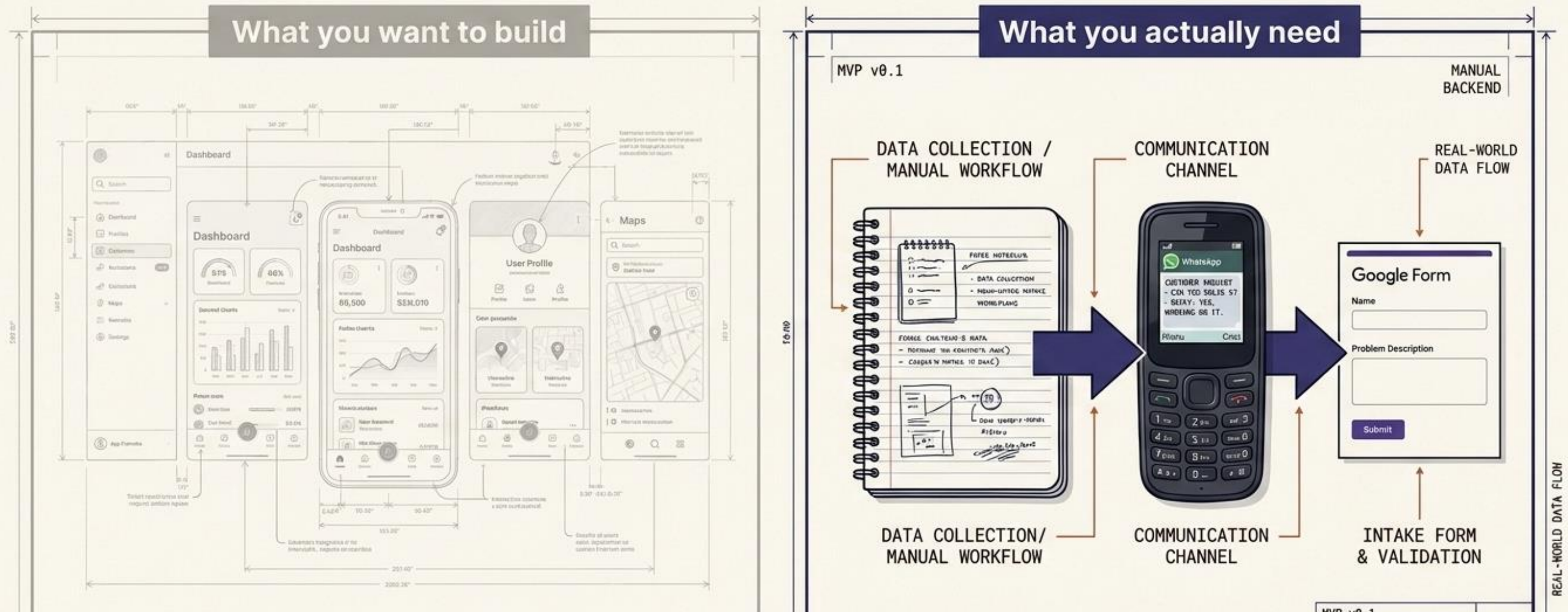
Execution Phase 1: The 20-Human Validation

Talk to 20 humans before you write a single line of code.



Execution Phase 2: The Ugly MVP

Your MVP should embarrass you.

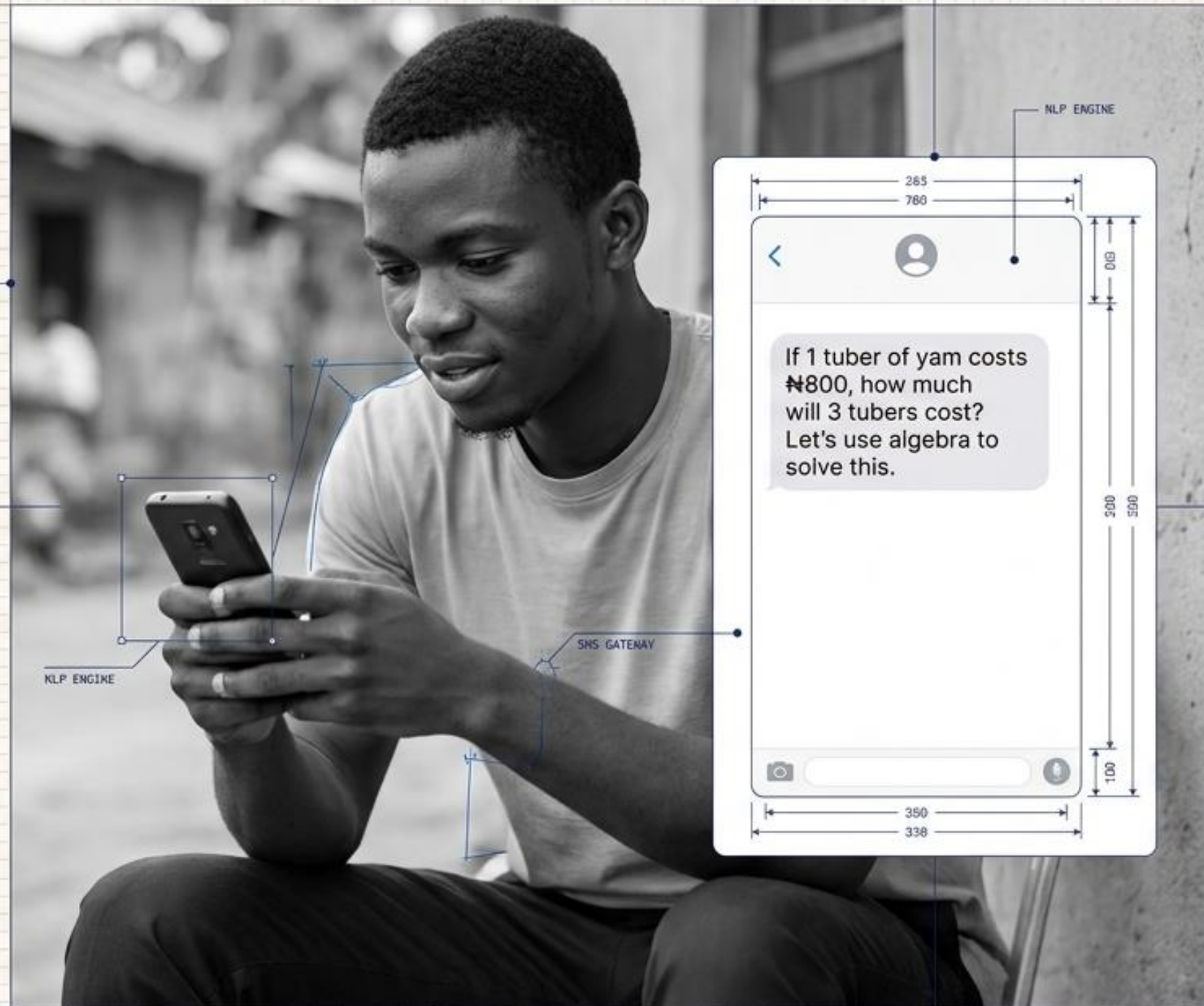


Sector Impact: AI in Education

Case Study: Rising Academies' Rori

150,000+ students

Impact delivered
at <\$5/year.



Sector Impact: AI in Healthcare

Case Study: RxAll

3M+ patients

Impact: Authentication served monthly.



CHEMICAL SPECTROSCOPY ANALYSIS
SIGNATURE: MATCH CONFIRMED



CLASS: ANTIMALARIAL
MODEL: RXALL AI
LATENCY: <500ms
ACCURACY: 99.8%

The Signal vs. Noise Matrix

What Actually Means Something.

NOISE

~~Great idea!~~

~~80% survey approval~~

~~Hackathon wins~~

~~Tech blog features~~

SIGNAL

A stranger pays twice

Organic referrals

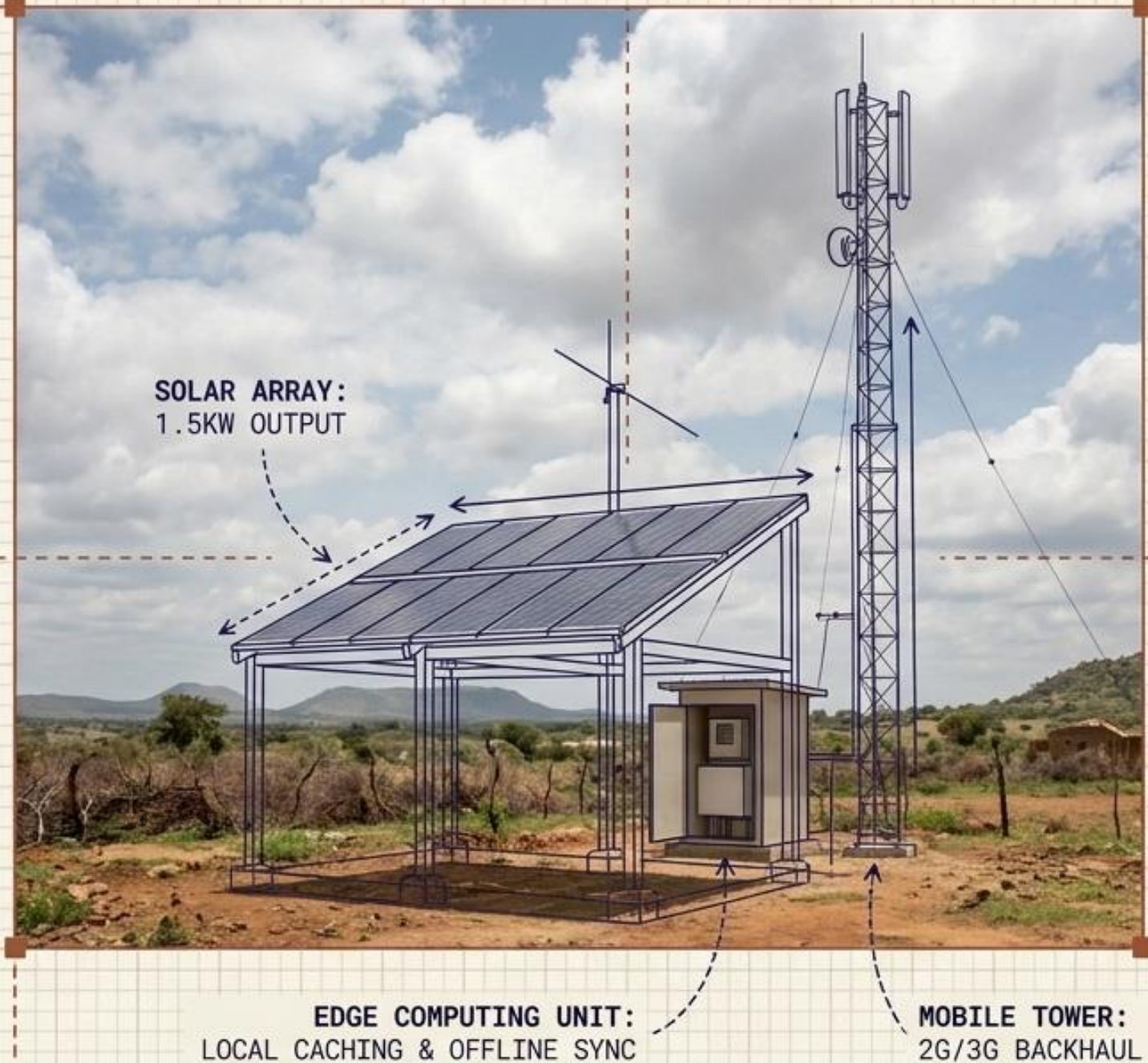
Flat retention curve

Anger when servers
are down

Scaling: Engineering for Reality

The Infrastructure Reality.

1. **Design Offline-First.**
2. **Local Caching.**
3. **Voice-First & USSD.**



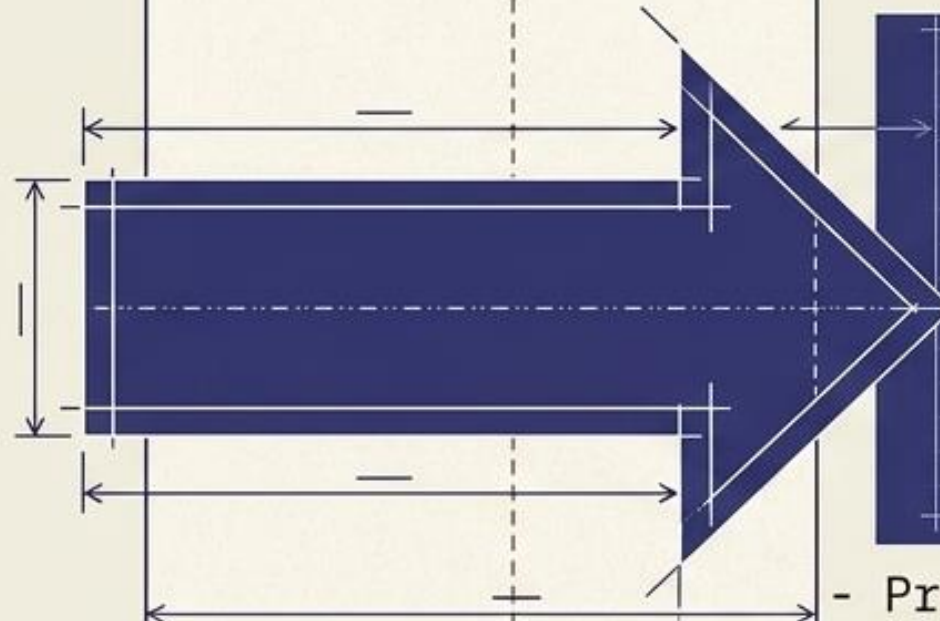
The Founder's Transition (Scale)

What got you here will kill you there.

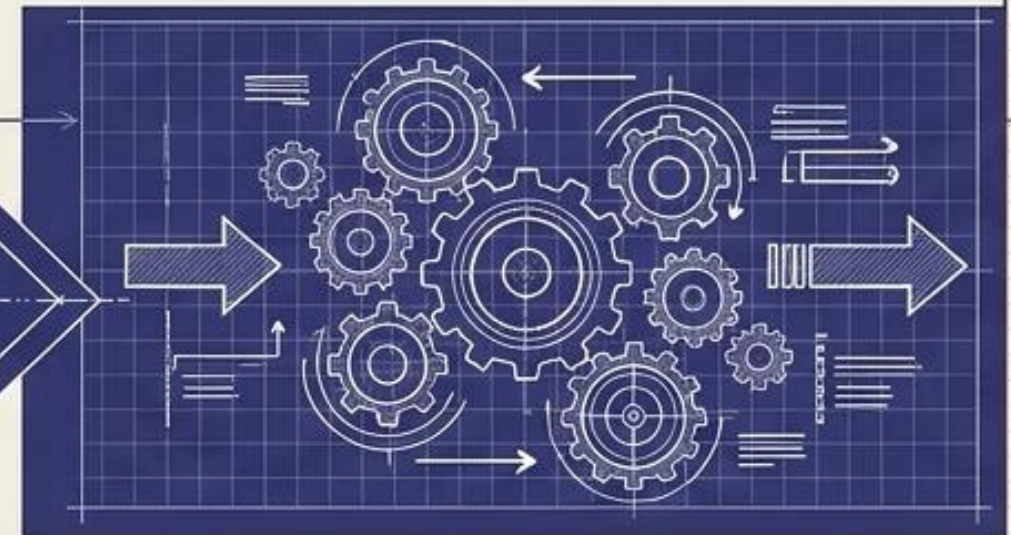
First 100 Users



- Speed.
- Founder does everything.
- WhatsApp support.



Next 10,000+ Users

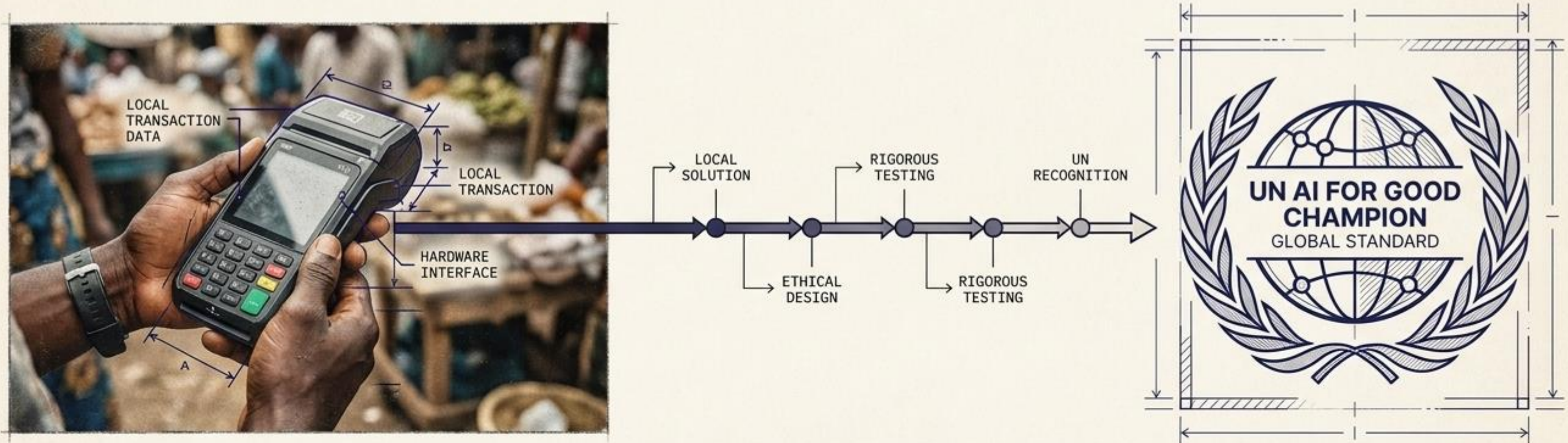


- Process.
- Founder does nothing operational.
- SLAs & Tickets.

Responsible AI & Global Impact

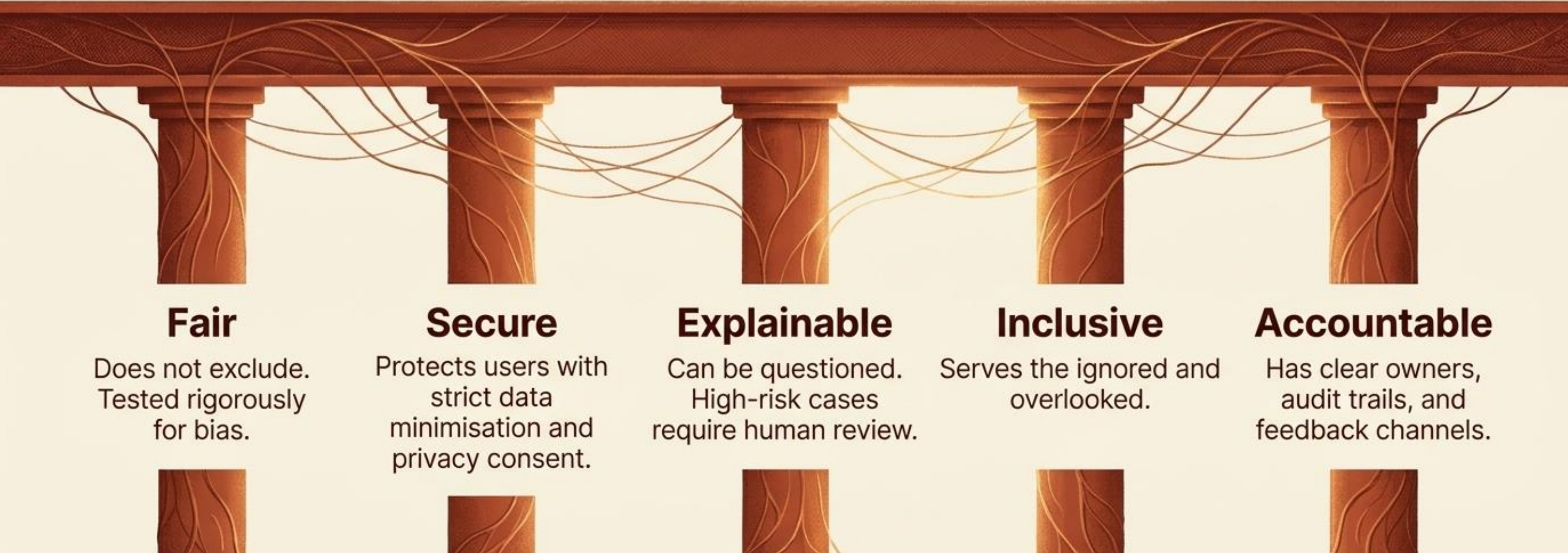
Case Study: Nearpays (UN AI for Good Champions)

From Local Pain to Global Standard.



Trust Must Be Designed In

AI can scale both benefit and harm. Responsible AI asks: who benefits, who may be harmed, who decides, and who is answerable?



Fair

Does not exclude.
Tested rigorously
for bias.

Secure

Protects users with
strict data
minimisation and
privacy consent.

Explainable

Can be questioned.
High-risk cases
require human review.

Inclusive

Serves the ignored and
overlooked.

Accountable

Has clear owners,
audit trails, and
feedback channels.

The Code to Impact Staircase

1. Code (Build)

Master programming, algorithms, ML evaluation, and deployment pipelines.

2. Prototype (Create)

Define the user's "job to be done." Design simple workflows. Validate usefulness with evidence, not just enthusiasm.

3. Product (Create)

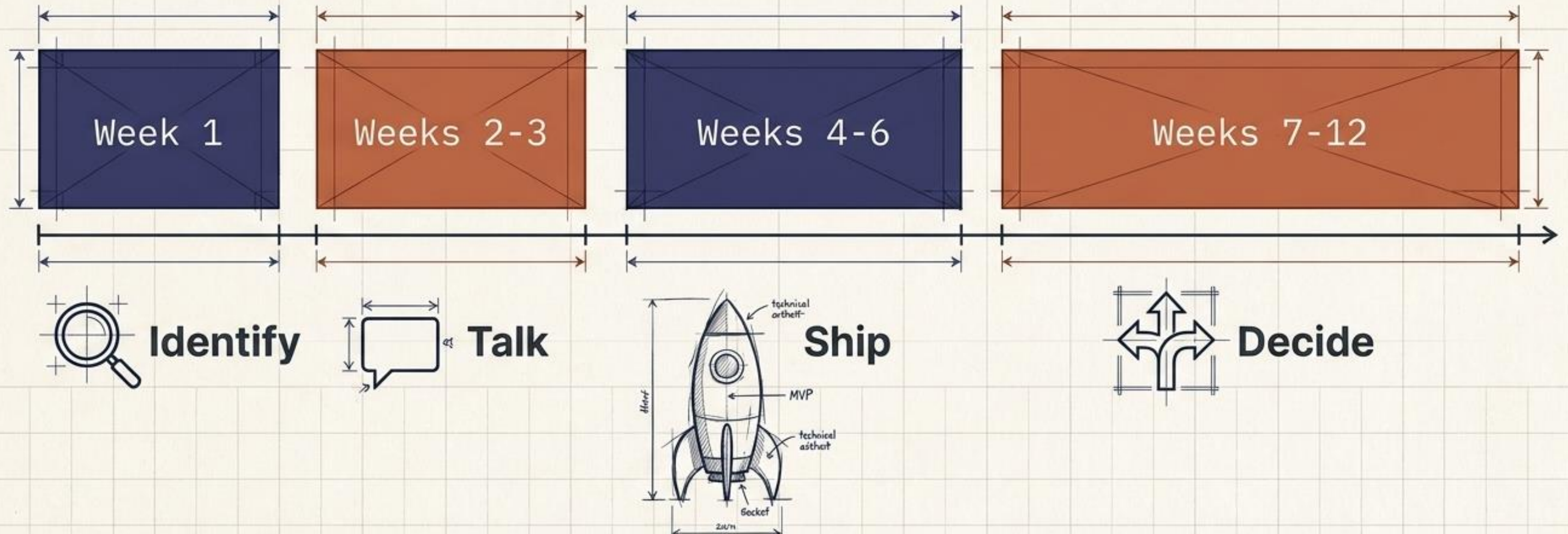
Build support, clear documentation, feedback loops, and maintenance plans.

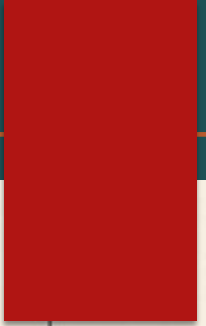
4. Global Scale (Scale)

Use modular architecture and APIs. Meet standards for security and privacy. Adapt across new markets and regulatory environments.

The 12-Week Semester Sprint

Your Turn. Do not wait for graduation.





Don't build to be a founder.
Build to be useful.

Useful gets paid. Useful compounds.



Build from where you are.

Create for the people
around you.

Scale with the world in mind.



Thank you for your attention.

NACOS CSCON 5.0

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